

Identification Problem

My paper looks at whether the 2018 Section 301 tariffs on Chinese imports caused U.S. multinational firms to shift more of their profits to foreign countries. The outcome variable is the effective tax rate (ETR), which goes down when firms move taxable income to low-tax countries through things like transfer pricing, intercompany licensing, or shifting debt across borders.

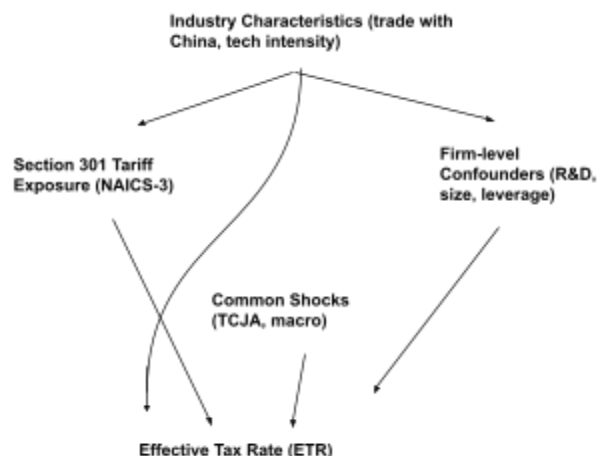
The main challenge is that tariff exposure wasn't random. The tariffs mostly hit tech-heavy industries like semiconductors, electronics, and machinery because the U.S. Trade Representative was targeting products tied to China's industrial policy. The problem is that these same industries already had a lot of offshore tax planning going on before the tariffs even existed. Tech firms own intellectual property like patents and software that is really easy to move on paper to a subsidiary in a low-tax country, while industries built around physical things like steel mills don't really have that option. So the industries that got hit hardest by tariffs were already the ones most likely to be booking profits abroad.

This creates a problem because if I just compare ETR changes between firms that were exposed to tariffs and firms that weren't, I can't tell if the tariffs actually caused more profit shifting or if those industries were just continuing what they were already doing. On top of that, the 2017 Tax Cuts and Jobs Act (TCJA) cut the corporate tax rate from 35% to 21% for everyone starting in 2018, which was right when the tariffs hit. So any simple before-and-after comparison would also pick up that unrelated policy change.

To get around this, I use a continuous difference-in-differences design that takes advantage of the fact that different industries were hit with different tariff rates, ranging from 10% to 21%. Instead of splitting firms into treated and control groups, I look at whether firms in industries with higher tariff rates saw bigger ETR drops after 2018 compared to firms in industries with lower rates. Firm fixed effects handle the fact that these industries were always different by comparing each firm to itself over time. Year fixed effects absorb the TCJA and other shocks that affected all firms at the same time. What's left is the effect of tariff intensity on ETR, which is the causal effect I'm trying to measure.

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The causal path goes from Section 301 tariff exposure to ETR, but there are some backdoor paths. First, industry characteristics create a backdoor path. Industries with deeper trade ties to China were more likely to be targeted by tariffs and may have already had different tax planning trends going on. This creates a non-causal link between tariff exposure and ETR. Second, common macro shocks like the TCJA, business cycle changes, and trade uncertainty hit all firms' ETR at the same time. Without controls, these get mixed into the tariff effect. Third, time-varying firm characteristics like size, R&D spending, and how much debt a firm has changed over time and independently affect both how a firm responds to tariffs and what its ETR looks like. Firm fixed effects close the backdoor through industry characteristics by comparing each firm to itself over time, and year fixed effects absorb common shocks including the TCJA. Time-varying controls address the firm-level confounders.



Key Model Equations

Continuous difference-in-differences with two-way fixed effects:

$$ETR_{it} = \beta(\text{TariffExposure}_j \times \text{Post2018}_t) + \gamma'X_{it} + \alpha_i + \delta_t + \varepsilon_{it}$$

ETR_{it} is the effective tax rate for firm i in year t

TariffExposure_j is the mean Section 301 tariff increase for the firm's NAICS 3-digit industry

Post2018_t equals 1 for 2019 onward since tariffs were imposed mid-2018

X_{it} includes time-varying controls

α_i are firm fixed effects, which control for everything about a firm that doesn't change over time

δ_t are year fixed effects, which control for anything that hit all firms in a given year like the TCJA

ε_{it} is the error term, clustered at the NAICS 3-digit level since that's where the tariff treatment varies

The coefficient β is what tells me whether firms with more tariff exposure saw a different change in ETR after 2018 compared to firms with less exposure. A negative β means more tariff-exposed firms had bigger ETR drops, which is consistent with profit shifting.

Event Study:

$$ETR_{it} = \sum_k \beta_k(\text{TariffExposure}_j \times 1[\text{Year} = k]) + \gamma'X_{it} + \alpha_i + \delta_t + \varepsilon_{it}$$

$\sum_k \beta_k$ are year-by-year coefficients for each year interacted with tariff exposure, with 2017 as the reference year

TariffExposure_j , X_{it} , α_i , δ_t , and ε_{it} are the same as the main model

The pre-treatment coefficients (β_{2015} , β_{2016}) test whether high-tariff and low-tariff firms were already on different paths before the tariffs happened. If those coefficients are close to zero, it supports the idea that the groups were trending similarly before 2018.

Core Assumptions

The key assumption is parallel trends, which is that firms in high-exposure and low-exposure industries would have followed similar ETR paths if the tariffs had never happened. The event study supports this since the pre-2018 coefficients are close to zero and not significant, and a placebo test with a fake 2017 treatment also comes back insignificant with the wrong sign.

Firms shouldn't have shown anticipation by changing their profit shifting behavior before the tariffs were actually imposed. I set the post indicator to 2019 to account for the lag between when tariffs hit mid-2018 and when it shows up in fiscal year financial statements.

One firm's tariff exposure shouldn't affect another firm's ETR through spillovers. Even if tariff-exposed firms pushed business to competitors, that would bias the estimate toward zero, making my results conservative.

The variation in tariff rates across industries needs to be exogenous to other things driving ETR trends. Section 301 tariffs were chosen based on China's technology transfer practices, not U.S. firms' tax behavior. Firm fixed effects absorb level differences across industries, and I address differential trends with industry-by-year fixed effects in robustness checks.

Threats to Identification

Industries hit by tariffs might have been on different ETR paths for reasons that have nothing to do with tariffs, like tech sectors already moving their tax structures offshore. I address this by adding industry-by-year fixed effects at different levels, from broad SIC 1-digit groupings up to narrower NAICS-2 groupings, and the effect holds across both.

Tariff intensity is measured at the NAICS 3-digit level, but firms in the same industry make different products with different actual exposure. This creates measurement error that biases the coefficient toward zero, so the estimated effect is a lower bound of the true effect.

The TCJA changed the corporate tax rate for everyone right around the same time. Year fixed effects absorb the parts of the TCJA that hit all firms equally, and since the tariffs were targeting trade with China and not tax behavior, there's no direct reason the TCJA would affect high-tariff and low-tariff industries differently.

The panel is unbalanced since firms enter and exit over time. Restricting to firms present in all 10 years gives nearly identical results, confirming that firms coming and going doesn't drive the finding.

Raw ETR has some extreme values from one-time tax events and data issues that could inflate the coefficient. The effect survives after capping at the top and bottom 5%, confirming it isn't purely driven by outliers.

Planned Robustness Checks

I run a series of robustness checks to make sure the main result holds up. To control for industry trends, I add industry-by-year fixed effects at different levels going from broad to narrow, and I run a placebo test with a fake 2017 treatment on pre-2019 data to check for pre-trends. I also re-estimate the model with tighter ETR winsorization and trimming, test foreign profit share as an alternative outcome, and try four different tariff exposure measures (mean rate, standard deviation, product counts, and variety counts). I restrict the sample to a balanced panel of firms present in all 10 years to check for attrition bias. Finally, since there are only 24 industries where the tariff treatment varies, I use a wild cluster bootstrap to make sure the standard errors aren't too small.